Keshav Sharma

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EDUCATION

Vellore Institute of Technology **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, Java, C/C++, SQL, JavaScript, R
Software Engineering: Data Structures & Algorithms, System Design, Distributed Systems, Operating Systems, Computer Networks, Database Management Systems, Multithreading & Concurrent Programming, Unit Testing, Test-Driven Development (TDD), Software Development Life Cycle (SDLC), REST APIs, Performance Optimization, Code Review
AI-Assisted Development: GitHub Copilot, Cursor, Claude, ChatGPT, Prompt Engineering, AI Coding Agents, Agentic AI Workflows, AI-assisted Debugging, Code Generation, Code Review
Backend & Frameworks: Flask, FastAPI, Node.js, React.js
Databases & Storage MySQL, SQLite, Redis (NoSQL), SQL Query Optimization
Tools & Platforms: Git, GitHub, GitHub Actions, Docker, PyTest, CI/CD, VS Code, IntelliJ IDEA, Jupyter Notebook

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** ***Aug – Sep, 2025

- Applied Julius AI and LLM-based prompt engineering to analyze 2 real-world datasets (8,500+ records), extracting 15 structured business insights through natural language querying and AI-assisted data analysis.
- Iteratively refined prompts and validated AI-generated outputs for consistency, accuracy, and relevance, improving the reliability of generated insights while demonstrating effective use of AI-assisted workflows.
- Delivered a capstone project recognized with AICTE, VOIS, and Vodafone Idea Foundation certification, presenting findings to 3 mentors and translating AI-driven analysis into 4 actionable recommendations on user engagement and content trends.

PROJECTS

***RedisLite* |LINK**

- Engineered RedisLite, a Redis-inspired in-memory key-value database in Python featuring a custom RESP protocol parser, multithreaded TCP server, thread-safe storage engine, and an O(1) LRU cache implemented using a hash map and doubly linked list, with lazy and active TTL expiration.
- Designed a modular four-layer architecture (TCP transport, storage engine, cache/TTL, persistence) implementing Write-Ahead Logging (WAL), atomic log compaction using os.replace(), automatic crash recovery, and concurrent request handling for reliable data persistence.
- Developed 46 automated PyTest unit tests following Test-Driven Development (TDD) principles, performed debugging and concurrency validation, and benchmarked 619K SET ops/sec for in-memory operations and 2.8K ops/sec with durable WAL persistence.

***API Rate Limiter Library* |LINK**

- Built a production-ready API rate-limiting library in Python implementing five industry-standard throttling algorithms, Redis integration, and reusable middleware for Flask and FastAPI applications.
- Engineered a modular architecture with storage abstraction, thread-safe concurrent request handling, configurable backends, and package distribution through PyPI for public installation and reuse.
- Implemented 30+ automated PyTest unit tests following Test-Driven Development (TDD) practices to validate edge cases, concurrency behavior, and algorithm correctness, ensuring production-grade reliability.

***Docwise* |LINK**

- Engineered Docwise, an AI-powered PDF Question Answering system using Google Gemini, enabling 95%+ query accuracy while reducing manual document search time by 40% through Retrieval-Augmented Generation (RAG) techniques.
- Implemented scalable document processing pipelines using TF-IDF vectorization, intelligent chunking, and real-time indexing for PDFs exceeding 200+ pages, improving responsiveness and retrieval performance.
- Applied prompt engineering, retrieval optimization, and hallucination-aware response validation to improve answer quality, while integrating chat history and session management in Streamlit to enhance usability and user retention by 35%.

Certifications

- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |LINK** **2026**
  Certified in Generative AI, LLMs, Prompt Engineering, RAG, AI Agents, and OCI AI services.
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate |LINK** **2026**
  Certified in AI, Machine Learning, Generative AI, LLMs, and OCI AI services.